Pathophysiology of Hemostasis

Yasir Eltahir

Introduction to Hemostasis

- Hemostasis prevents and stops bleeding.

- - Involves vascular spasm, platelet plug formation, coagulation, and fibrinolysis.

- Imbalance leads to excessive bleeding or thrombosis.

Key Components of Hemostasis

- Platelets: Form a plug at the injury site.
- Clotting Factors: Help in blood clot formation.
- Vascular Endothelium: Regulates clotting.
- Fibrinolytic System: Prevents excessive clotting.

Thrombocytopenia

- Low platelet count (<150,000/µL).
- Causes: Bone marrow suppression, autoimmune diseases, infections.
- Symptoms: Easy bruising, prolonged bleeding.
- Nursing: Bleeding precautions, monitor platelets.

Von Willebrand Disease (vWD)

- Genetic disorder affecting platelet adhesion.

- Types: Type 1 (mild), Type 2 (moderate), Type 3 (severe).

- Symptoms: Frequent nosebleeds, heavy menstrual bleeding.

- Management: Desmopressin, vWF therapy, avoid NSAIDs.

Hemophilia

- X-linked disorder (Factor VIII or IX deficiency).
- Symptoms: Joint/muscle bleeding, prolonged bleeding.
- Management: Factor replacement therapy.
- Nursing: Monitor bleeding, educate on injury prevention.

Disseminated Intravascular Coagulation (DIC)

- Widespread clotting leads to excessive bleeding.

- Causes: Sepsis, trauma, cancer, obstetric complications.

- Symptoms: Uncontrolled bleeding, organ damage.

- Management: Treat cause, transfusions, anticoagulants.